Project 1 - Explainable AI in Radiology

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Abstract

This is a great paper, and it has a concise abstract.

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1. Introduction

Machine learning has become a powerful tool for solving complex problems in various domains and is increasingly integrated into different applications. However, many of these models act as black boxes with limited knowledge of how they work and what led to a certain output(Ahmed et al., 2022) . Explainable AI (XAI) aims to improve the understanding of these systems by making the decision process more transparent. Healthcare, specifically radiology imaging, is one area with increased usage of AI systems to support diagnostic and treatment decisions(E. Ihongbe et al., 2024) . To ensure trust in these systems, especially when crucial medical decisions must be made with their help, it becomes increasingly important to not only produce efficient and accurate models but also explainable ones to increase trust in the model's outputs. Many AI tools in medicine are meant to support medical personnel in their decision processes and decrease their workload. Explainable models will increase the personnel's confidence in the decisions made by the system and let them detect any potential errors. One of the current main applications for AI in medicine is the analysis of X-ray images to detect various illnesses, most often utilized for bone or lung diseases (Miró-Nicolau et al., 2022) . Chest Radiography (CXR) images are one example of this technique. With many illnesses affecting the chest area and lungs, accurate systems are vital. They have become more relevant since the COVID-19 outbreak(Miró-Nicolau et al., 2022) , and people have not only realized how important good systems are but also the importance of being able to understand the system's decision through explainability. In this paper we will present the application of machine learning to X-ray images for predicting pleural effusion (a buildup of excess fluid in the lungs) and two explanation methods: Saliency maps and Cosine Similarity. Additionally, these methods are evaluated to verify their performance using Quantus(Hedström et al., 2023) and (—–).

2. Methods & Materials

Dataset and Preprocessing

The dataset used for this study was a downsized version of the CheXpert dataset cite, which contains both frontal and lateral chest X-rays. Among the available pathology labels, particular attention was given to the *Pleural Effusion* label, which served as the target for classification.

A series of preprocessing steps were applied to prepare the dataset for model training. Following is the pipeline that it was followed.

1. The dataset was filtered to have only frontal chest X-ray images, excluding lateral views to ensure consistency.

2. All images were resized to a uniform resolution of (224×224) pixels, to facilitate compatibility with common CNN architectures. Moreover, pixel values were normalized to the range $[0, 1]$, ensuring consistent input scaling.

3. Images where the *Pleural Effusion* label was either missing or marked as uncertain were removed in order to improve reliability.

4. Since the number of images with positive Pleural Effusion labels was considerably lower than the number of negatives, class weights were computed based on the relative frequencies of each class. [1]

5. The dataset was split into training and validation subsets. Specifically, 80% of the images were allocated to the training set, while the remaining 20% formed the validation set.

6. Since the original chest X-rays were grayscale, each image was duplicated across three channels to match the expected input format for models pretrained on ImageNet, which typically expect 3-channel RGB images.

2.1. Model Architecture: DenseNet169

To develop the classification model for detecting *Pleural Effusion*, it was adopted a deep learning approach using **DenseNet169**, a convolutional neural network (CNN) architecture that belongs to the DenseNet family, originally introduced by Huang et al. in 2017 cite.

DenseNet, short for *Densely Connected Convolutional Network*, is distinguished by its unique connectivity pattern: every layer is directly connected to every other layer in a feed-forward fashion. Specifically, the output feature maps of each layer are concatenated with the function of all preceding layers and passed as input to subsequent ones. This dense connectivity pattern offers several advantages, such as: feature propagation and efficient parameter use.

DenseNet169 is a specific configuration within the DenseNet family, consisting of 169 layers, including convolutional layers, batch normalization layers, and ReLU activation functions, grouped into several densely connected blocks. Between these blocks are *transition*

1. These weights were later used during model training to help mitigate the imbalance.

layers, which downsample the spatial dimensions using pooling operations, while also reducing the number of channels to improve computational efficiency.

DenseNet169 strikes a balance between depth and computational complexity, making it particularly well-suited for medical imaging tasks, where subtle patterns often require deep networks with strong feature propagation capabilities.

The model was initialized with pre-trained weights from the ImageNet dataset. This pre training provides the model with a rich set of low- and mid-level visual features (such as edges, textures, and shapes), which are also relevant for analyzing medical images. During the initial stages of training, the pre-trained layers were frozen to preserve these general features. Fine-tuning was then performed on the higher layers, allowing the network to adapt specifically to the characteristics of chest X-rays and the task of detecting *Pleural Effusion*.

2.2. Saliency Map-based Explanations

One form of visual explanation for deep learning models is saliency map-based explanations. In this report, we specifically explore the use of Gradient-weighted Class Activation Mapping (Grad-CAM), first introduced by Selvaraju et al., which uses gradients in the last convolutional layer of the CNN and global-average-pooling to determine each neuron's importance to the final prediction(Selvaraju et al., 2017), and can be used to generate heatmaps to highlight the critical information in the image. Grad-CAM has often been used in medical image analysis with good results, for example for classifying tuberculosis lung zones(Devasia et al., 2023), identifying Covid-19 in chest X-rays(Khan et al., 2022) and pneumonia(Du and Yang, 2023). Additionally, we will use Quantus(Hedström et al., 2023), a toolkit for evaluating deep learning explanations, to evaluate the quality of the generated explanations. Specifically, we will be using the following metrics: Pixel-Flipping(Bach et al., 2015) for Faithfulness, Local Lipschitz Estimate(Alvarez Melis and Jaakkola, 2018) for Robustness, Model Parameter Randomisation test(Adebayo et al., 2018) for Randomisation, Relevance Mass Accuracy(Arras et al., 2022) for Localisation, Complexity(Bhatt et al., 2020) for Complexity

2.3. Example-based Explanations

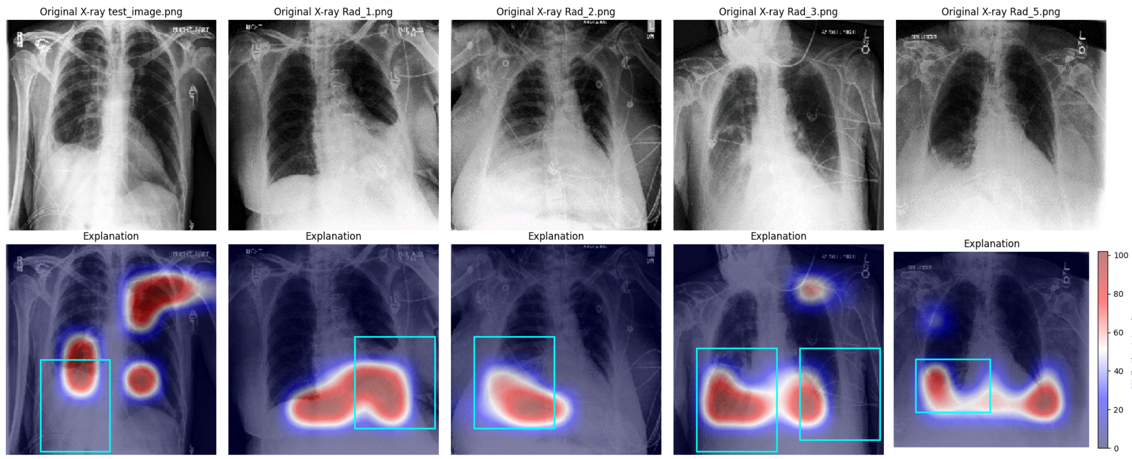
3. Results & Discussion

In the subsequent sections we will present and discuss the results gathered from testing the model and explanation methods described in section2 on a predefined set of test images.

Saliency Maps

The GradCAM explanations can be seen in figure1. As seen in the colour, the most important regions in the image, as determined by the model, are portrait in seismic colours: red, then white and lastly blue. The bounding boxes on the explanation boxes circle the important parts of the image, that the explanation should focus on. As seen in the five test images, the given explanations always highlight some part of the bounding box, but in most cases also falsely highlights parts of the X-ray image outside the box.

Figure 1: GradCAM Explanations



Additionally, in table1 we can see the mean and standard deviation of the quantus evaluations metrics on the test images. The complexity metric demonstrates how easily understandable the explanations are and the value of 9.3 is as expected for a complicated model such as a CNN. For the robustness measure, a low value, such as 0.032051 indicates a very robust model, that is stable to slight perturbations of the given input. Lastly, the faithfulness measure shows ...

Table 1: Quantus: Explanation Evaluation

	Mean	**Standard deviation**
Complexity	9.360562	0.241967
Local Lipschitz Estimate	0.032051	0.031284
Pixel Flipping	1	0

4. Model Training and Evaluation

During the training process, the DenseNet169 model demonstrated steady improvement in both training accuracy and training loss across epochs, indicating that the model was progressively learning useful features from the training data. However, the validation accuracy fluctuated across epochs, suggesting some degree of overfitting (Table 2).

Despite these fluctuations in validation accuracy, the final trained model demonstrated **reasonable performance** on the held-out test set.

Specifically, the model achieved an **overall accuracy** of 79.21%. In terms of class-specific performance, the model obtained a precision of 63.41% and a recall of 81.25%, reflecting the model's relatively good ability to correctly identify positive *Pleural Effusion* cases.

To furthermore a confusion matrix was computed (Image **??**).

Table 2: Training and Validation Performance Per Epoch

Epoch	Training Accuracy	Training Loss	Validation Accuracy	Validation Loss
1	80.40%	0.4420	83.85%	0.3845
2	83.57%	0.3936	85.59%	0.3497
3	83.97%	0.3836	84.89%	0.3601
4	84.39%	0.3747	82.55%	0.3992
5	84.51%	0.3697	80.96%	0.4323
6	84.76%	0.3613	85.83%	0.3470
7	84.99%	0.3581	85.09%	0.3595
8	85.37%	0.3494	85.48%	0.3530
9	85.65%	0.3436	82.36%	0.4055
10	85.89%	0.3364	84.75%	0.3746

The F1-score reached 71.23%, indicating a balanced trade-off between precision and recall. Furthermore, the model achieved a ROC AUC score of 84.96%, confirming good overall discriminatory power in distinguishing between positive and negative cases.

To statistically assess the significance of these results, a binomial test was conducted. *explain better the test and the threshold* The resulting p-value of 2.03×10^{-17} strongly supports the conclusion that the model's performance is significantly better than chance.

4.1. Example-based Explanations

5. Conclusion

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References

Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. *Advances in neural information processing systems*, 31, 2018.

Saad Bin Ahmed, Roberto Solis-Oba, and Lucian Ilie. Explainable-ai in automated medical report generation using chest x-ray images. *Applied Sciences*, 12(22):11750, 2022.

David Alvarez Melis and Tommi Jaakkola. Towards robust interpretability with self-explaining neural networks. *Advances in neural information processing systems*, 31, 2018.

Leila Arras, Ahmed Osman, and Wojciech Samek. Clevr-xai: A benchmark dataset for the ground truth evaluation of neural network explanations. *Information Fusion*, 81:14–40, May 2022. ISSN 1566-2535. doi: 10.1016/j.inffus.2021.11.008. URL http://dx.doi.org/10.1016/j.inffus.2021.11.008.

Sebastian Bach, Alexander Binder, Grégoire Montavon, Frederick Klauschen, Klaus-Robert Müller, and Wojciech Samek. On pixel-wise explanations for non-linear classifier decisions by layer-wise relevance propagation. *PloS one*, 10(7):e0130140, 2015.

Umang Bhatt, Adrian Weller, and José MF Moura. Evaluating and aggregating feature-based model explanations. *arXiv preprint arXiv:2005.00631*, 2020.

James Devasia, Hridayanand Goswami, Subitha Lakshminarayanan, Manju Rajaram, and Subathra Adithan. Deep learning classification of active tuberculosis lung zones wise manifestations using chest x-rays: a multi label approach. *Scientific Reports*, 13(1):887, 2023.

Jiajie Du and Jiaqi Yang. Classification of chest x-ray images (pneumonia) based on resnet and grad-cam. In *Proceedings of the 2023 6th International Conference on Signal Processing and Machine Learning*, pages 156–164, 2023.

Izegbua E. Ihongbe, Shereen Fouad, Taha F. Mahmoud, Arvind Rajasekaran, and Bahadar Bhatia. Evaluating explainable artificial intelligence (xai) techniques in chest radiology imaging through a human-centered lens. *Plos one*, 19(10):e0308758, 2024.

Anna Hedström, Leander Weber, Daniel Krakowczyk, Dilyara Bareeva, Franz Motzkus, Wojciech Samek, Sebastian Lapuschkin, and Marina M-C Höhne. Quantus: An explainable ai toolkit for responsible evaluation of neural network explanations and beyond. *Journal of Machine Learning Research*, 24(34):1–11, 2023.

Muhammad Attique Khan, Marium Azhar, Kainat Ibrar, Abdullah Alqahtani, Shtwai Alsubai, Adel Binbusayyis, Ye Jin Kim, and Byoungchol Chang. Covid-19 classification from chest x-ray images: a framework of deep explainable artificial intelligence. *Computational Intelligence and Neuroscience*, 2022(1):4254631, 2022.

Miquel Miró-Nicolau, Gabriel Moyà-Alcover, and Antoni Jaume-i Capó. Evaluating explainable artificial intelligence for x-ray image analysis. *Applied Sciences*, 12(9):4459, 2022.

Ramprasaath R Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE international conference on computer vision*, pages 618–626, 2017.